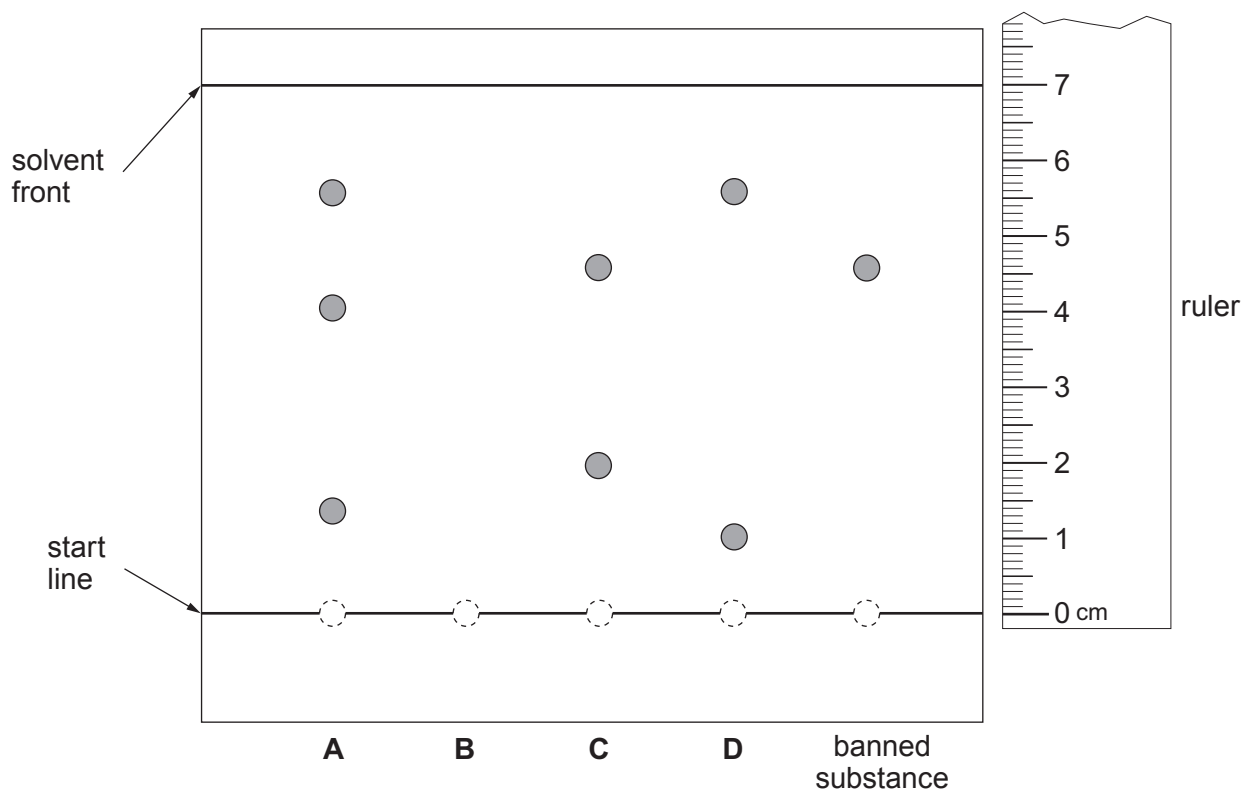


2. A food company was accused of using a banned substance in its sweets. Scientists tested four dyes, **A-D**, to find out if this was true or not. The results are shown below.



- (a) Name the method used.

[1]

.....

- (b) Dye **B** was found to have an R_f value of 0.428.

- (i) Use the following equation to calculate the distance moved by dye **B**.

[2]

$$\text{distance moved by dye B} = R_f \times \text{distance moved by solvent front}$$

Distance moved = cm

- (ii) **Complete the diagram** above to show the position of dye **B**.

[1]



(c) Identify which dye, **A-D**, contains the banned substance. Give a reason for your answer. [2]

Dye

Reason

.....

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05

